

Week 4 Write Up for
Master of Science
Information and Communications Technology

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Overview

The main challenge addressed is monitoring visitors to different parking lots and billing them monthly based on their parking duration. The system's features include registering customers and their vehicles, as well as recording parking locations and times. This week, we developed a system layer dedicated to registering cars and customers.

Because I started this quarter with the given netbeans codebase, the Address class already had the Builder pattern set up. And since I needed to research it during week 1 in order to use it correctly in my testing and implementation, I found it very simple to implement the same pattern in the Customer and ParkingTransaction classes.

Challenges Faced

As mentioned, I didn't find this assignment too challenging, the only real thing that I had to work around was my charge creation logic. I originally calculated the Money to associate with the ParkingCharge in the ParkingCharge constructor. But I realized that the logic should really be inside of the park method of the TransactionManager class. So, I kept the logic intact but moved it over to the correct method. Once I did that, I ran all current JUnit tests that I had already created to make sure that any refactoring I did didn't cause any new errors or unexpected results. Doing this, I found that any of my tests that had a Customer object created needed to be updated to use the new Builder instead of the constructor method.

JUnit Testing

I setup the JUnits for the ParkingChargeStrategyFactory by checking that the getStrategyName was correct. As for the TransactionManagerTest, I had some trouble with deciding how to check if the park method was working correctly. The way I ended up doing it

was to create the parking office, parking lot, add the lot to the office, and register both the car and customer to the parking office. Then I used the park method and also used the ParkingTransaction constructor to make the Transaction. Then I compared the get methods to make sure they had the same values. I also decided to add in JUnit tests for the Customer and ParkingTransaction classes, even though they weren't officially part of the assignment. I refactored them for the discussions, so I wanted to verify that they were working as I expected.

The Customer and ParkingTransaction classes didn't have a JUnit test file made yet, so I also did have to create files for them and test the getters and setters for each. This was a bit tedious but not too difficult.

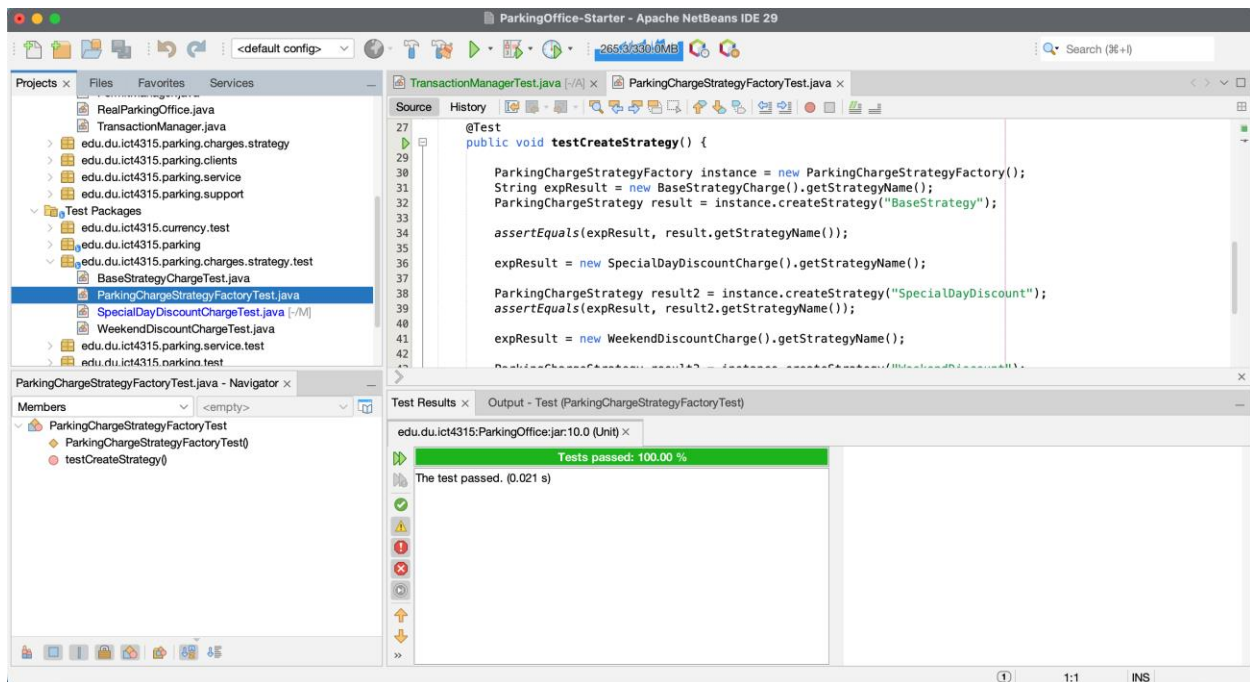


Figure 1. JUnit test results for ParkingChargeStrategyTest

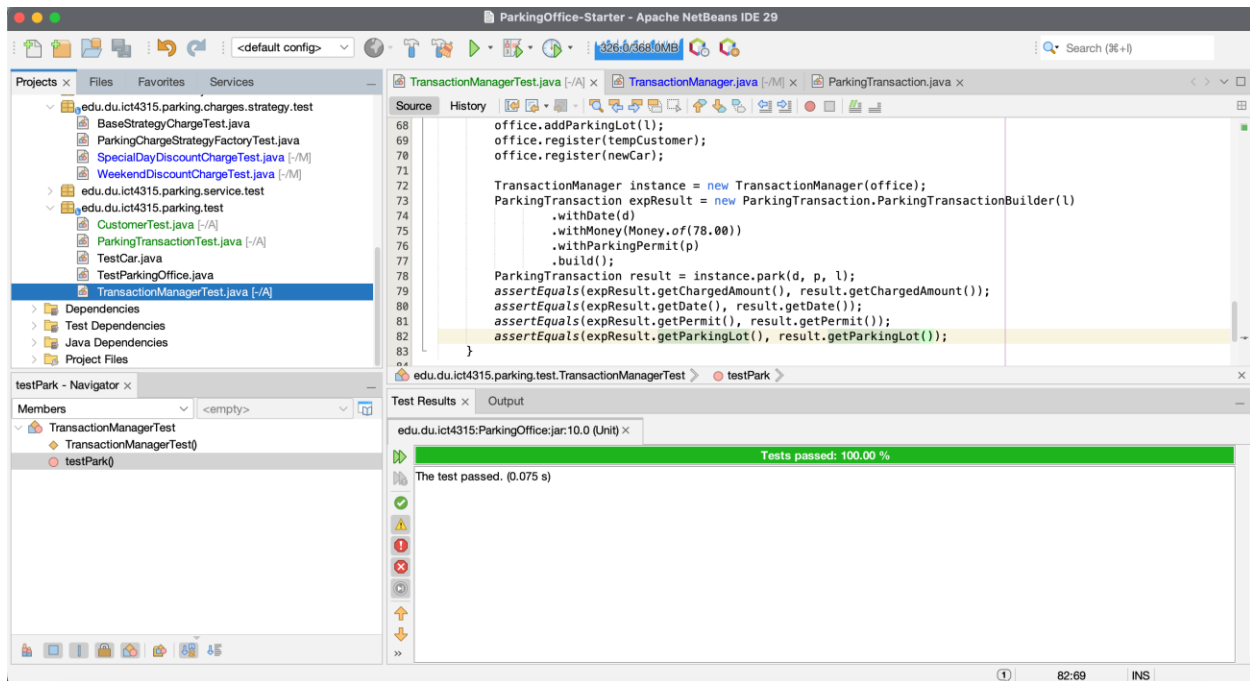


Figure 2. JUnit test results for TransactionManagerTest

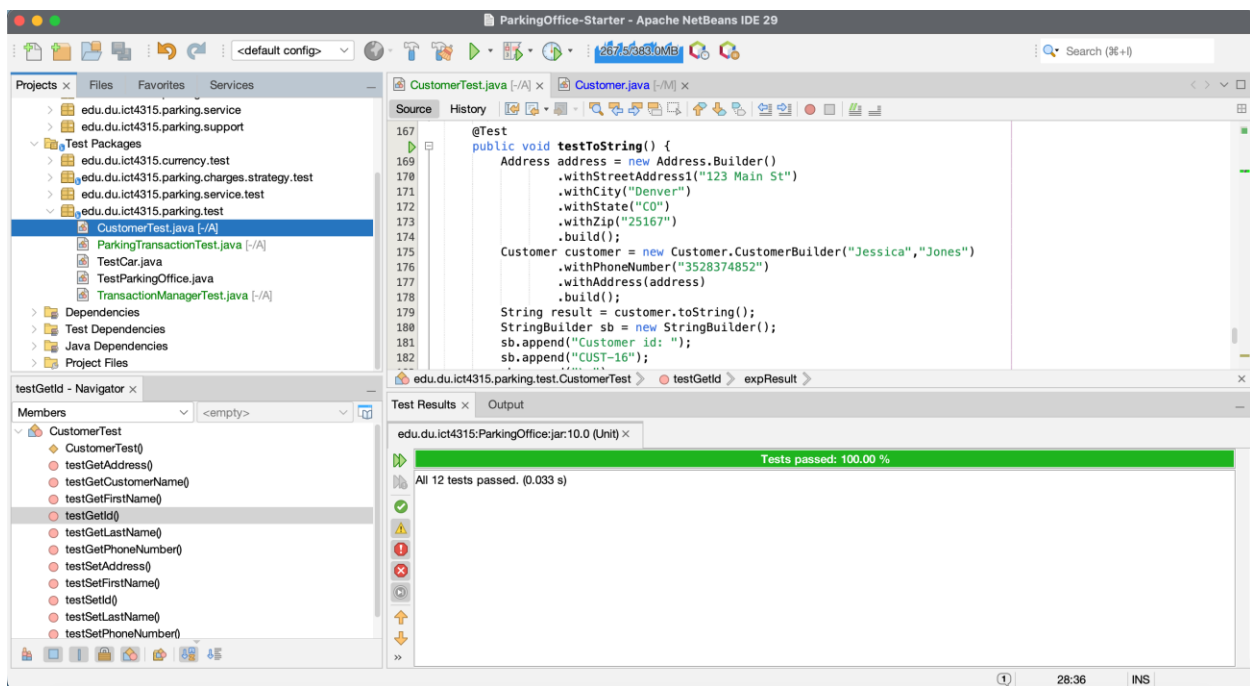


Figure 3. JUnit test results for CustomerTest

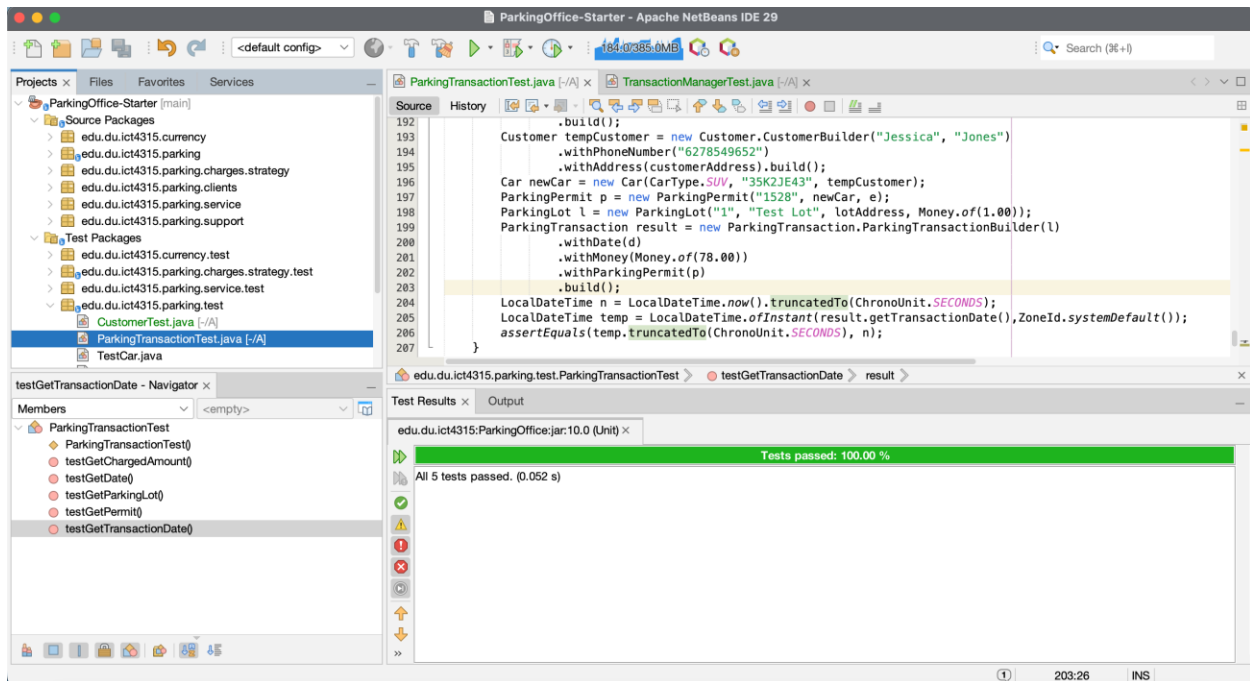


Figure 3. JUnit test results for ParkingTransactionTest

Future Considerations

Moving forward, I want to dedicate some time to reevaluating the charging logic I'm currently using. Although the calculations appear correct, I'm not confident the system is functioning as intended. I recall that certain parking lots used daily rates while others applied hourly rates, and I suspect my current logic doesn't properly account for both scenarios.

As noted in my previous reports, I intend to utilise the Logger library exclusively for handling issues, rather than employing both Logger and exception handling mechanisms. This week did not require exception handling; therefore, while I began studying how to implement Logger, I have not yet incorporated it into my code.